

Software Project Management (SPM) (CSC415)

Course Instructor
Er. ANUJ GHIMIRE

DISCLAIMER

- *This document does not claim any originality and cannot be used as a substitute for prescribed textbooks.*
- *The information presented here is merely a collection from various sources as well as freely available material from internet and textbooks.*
- *The ownership of the information lies with the respective authors or institutions.*



Chapter 7: Managing Contract and People

Introduction (Contract)

- Outsourcing is often viewed as involving the contracting out of a business function - commonly one previously performed in-house - to an external provider.
- In this sense, two organizations may enter into a contractual agreement involving an exchange of services and payments.
- A contract is an agreement between two parties that creates an obligation to perform (or not perform) a particular duty.
- It is a two-party agreement, either given verbally or in writing, that provides a product or service to an individual or business.

Contract

- Consideration between the parties must be exchanged for the contract to be lawful and enforceable.
- The purpose of a contract is to protect the parties if either party does not fulfill its promises.
- If the contract is broken, it can be turned over to a court for enforcement. *The remedy at law for breach of contract is “damages” or monetary compensation.*
- **Every great project starts with a contract.**

Contract

- Contracts can be done through:
 - *Letter of Agreement (LOA),*
 - *Memorandum of Understanding (MOU),*
 - *Letter of Intent (LOI), or*
 - *Any other name, if it contains all the referenced elements, then it's a contract.*

Types of Contract

- The available options are the following, and we need to understand their differences to know the outcome of the software development process
- Contract types for project management:
 - *Fixed-Price Contract (FP)*
 - *Cost Reimbursable Contract (CR)*
 - *Time and Material Contract (T&M)*
 - *Software Development Subscription Contract (SDS)*

Types of Contract (*Fixed-Price Contract*)

- Project managers prefer using fixed-price contracts when the payment does not depend upon resources or time expended on completing the project.
- In a fixed-price contract, the seller estimates the total cost of labor, resources and materials and completes the action regardless of the actual cost.
- In such contracts, buyers carry less risk and the seller typically receives their fixed price for finishing the project.
- The construction industry widely uses fixed-price contracts.

Types of Contract (*Fixed-Price Contract*)

- With this type of contract, buyers know the actual project cost, resulting in minimal buyer risk.
- At the start of the project, a buyer makes some payment and the seller takes most of the risk because the buyer pays only for work upon completion.
- Some types of fixed-price contracts include:
 - *Fixed-Price Incentive Fee Contract*
 - *Firm Fixed-Price Contract*
 - *Fixed-Price With Economic Price Adjustments Contract*

Types of Contract (*Cost Reimbursable*)

- This type is used when the requirements are uncertain from one side and the development process is not clear from the other.
- It is used for new research and development and requires immense innovation without a guarantee of predicted outcome.
- The key idea of this contract is that the seller provides work for a fixed time period and then increases the bill to get profit after finishing the product.

Types of Contract (*Cost Reimbursable*)

- The amount of the raise in this case is unknown to the other party, because it is not discussed prior to the agreement.
- There are subtypes that you can choose from if you decide to go for this option:
 - *Cost Plus Percentage of Cost*
 - *Cost Plus Fixed Fee*
 - *Cost Plus Incentive Fee*
 - *Cost Plus Award Fee*

Types of Contract *(Time & Material Contract)*

- This is the second popular option after Fixed Price Contract, and it is a hybrid of both Fixed Price and Cost Reimbursable.
- One of the parties agrees to pay the other the time and materials that are used for the project within a reasonable limit.
- It can be cost reimbursable when the customer agrees to pay the cost for all the genuine and legitimate expenses. Or, it can be more like a FP type when the customer sets the limit.

Types of Contract *(Time & Material Contract)*

- In this case, the vendor is selected based on the capabilities and experience, having the required manpower and materials.
- The cost for supplies is negotiated and is paid according to the quantity of the resources consumed or purchased.
- The contract is pretty simple and convenient for both parties, and it is possible to establish a not-to-exceed price to avoid massive cost overruns.

Types of Contract (*SDS Contract*)

- SDS stands for Software Development Subscription Contract.
- A software subscription agreement is a contract between a company and its customer that allows the customer to use software for an agreed-upon amount of time, often with updates or upgrades included.
- A typical software subscription agreement consists of the type of licensing rights granted, when the license starts and ends, how many licenses will be issued, and whether any restrictions apply.

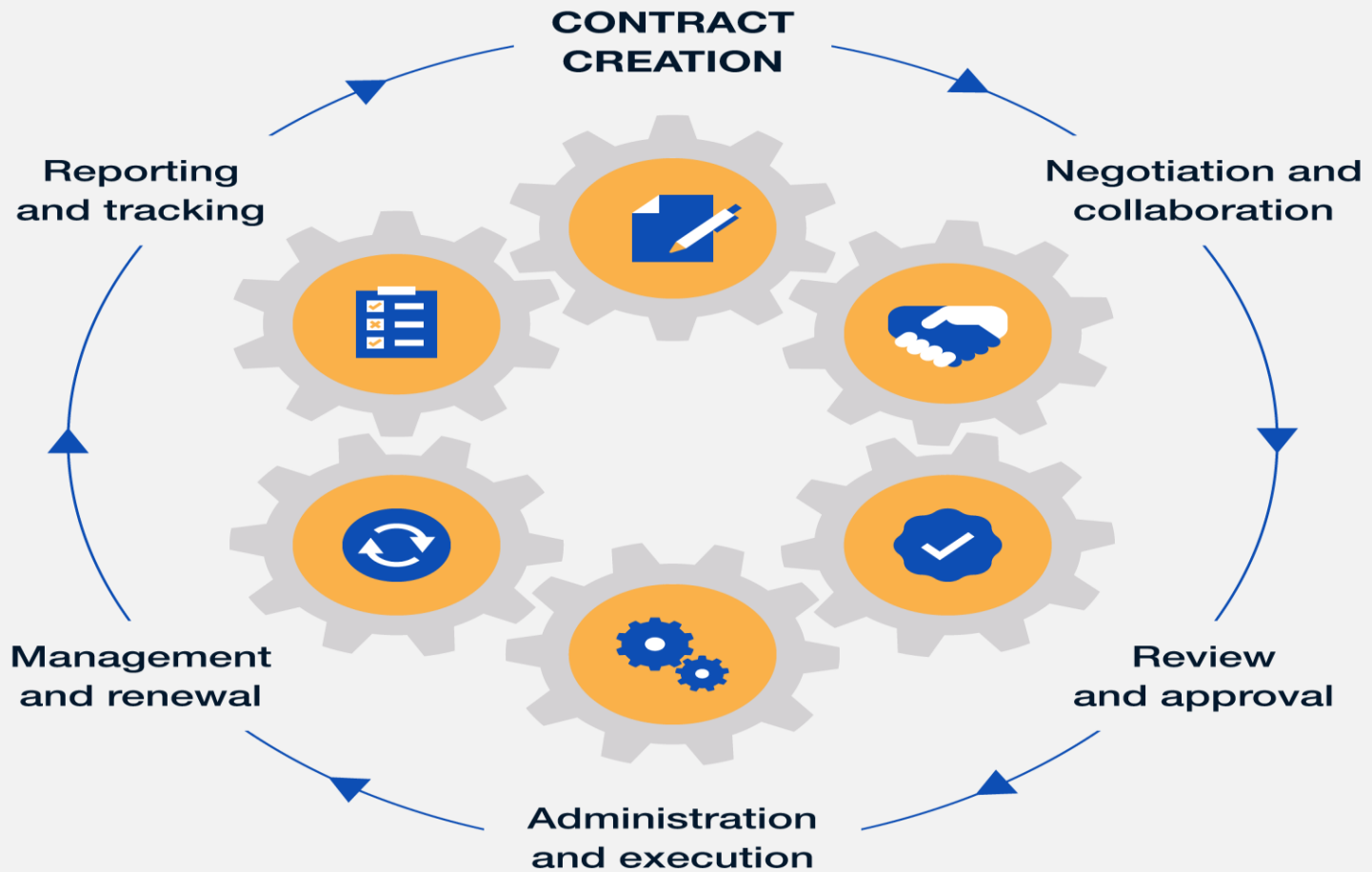
Types of Contract (*SDS Contract*)

- Software subscriptions may also include maintenance agreements where companies provide bug fixes and security patches.
- In addition, some software providers offer free trials so customers can try out products before making their purchase decision, while others provide discounts on upgrading from one version to another.

Stages in Contract

- The Contract Management Lifecycle (CML) involves everything from creating your contract to termination and tracking.
- This cycle can be broken down into the six distinct stages of contract management.
- The prominent stages of a contract lifecycle are:
 - *Step_1: Contract Creation*
 - *Step_2: Negotiation and Collaboration*
 - *Step_3: Review and Approval*
 - *Step_4: Administration and Execution*
 - *Step_5: Ongoing Management and Renewal*
 - *Step_6: Reporting and Tracking*

Contract Lifecycle



Stages in Contract

- Steps one to three occur before the contract is signed (pre-signature) and four to six occur post-signature.
 - **Pre-signature activities** involve everything from creating the contract to signing on the dotted line.
 - **Post-signature activities** occur after the contract is signed and continue until it's either renewed or terminated.
- Contract Lifecycle Management (CLM) makes the process more intuitive so you can keep track of your contracts at every stage in the contract lifecycle.

Stages in Contract (Creation)

- After you get a verbal agreement, the next step is to get it in writing.
- In many cases, you'll be working off a template for the contract, but it still requires careful attention to personalize the contract for each agreement.
- Trigger events for contract creation include:
 - *Verbal Agreements*
 - *New Contract Requests or Intake Requests*
 - *Contract Amendments*
 - *Contract Renewals*
 - *Contract Cancellations*

Stages in Contract (Creation)

- When you're creating a contract, it's important to make sure it covers all the relevant information about the agreement.
- Some of relevant information regarding agreement can be:
 - *Services each party will Receive or Provide*
 - *Start Dates*
 - *Pricing*
 - *Projected Milestones and Expectations*
 - *Contingencies*

Stages in Contract (Negotiation and Collaboration)

- Contract negotiation is one of the most important steps in the contract lifecycle.
- After a contract is generated, all parties involved, need to negotiate back and forth until final terms are agreed to.
- This stage in the process is the most time-intensive.
- Standard steps such as collecting signatures, catching mistakes, and legal approval are the biggest obstacles for faster negotiation.

Stages in Contract (Negotiation and Collaboration)

- Keeping the negotiation and process centralized through online the collaboration tools reduces the possibility of human error and ensures all parties have access to the most recent version of the contract.
- These features give you access to a contract at each stage, so the right users have full visibility into the negotiation process.

Stages in Contract (Review and Approval)

- Once negotiation and required collaboration generates contract draft, we need to send it up the approval chain for each internal party to review.
- These approval chains typically include managers, officers, and other stakeholders.
- The types of roles and number of people who are looped in for approval will vary widely depending on the organization.
- This is a stage where many contracts get stuck in dividing line, delays occur as people lose track of the contract, especially if late-stage edits occur.

Stages in Contract (Administration & Execution)

- Once a contract's been approved, it's time to finally put pen to paper.
- Although many team members are involved in contract review and approval, there's typically only one person from each party who needs to sign on the dotted line.
- Once everyone's signed, the contract gets securely encrypted and stored in a contract repository for secured storage.
- Then both the parties continues to work as per the agreement made through the contacts

Stages in Contract (Management & Renewal)

- Contract doesn't stop once you've signed on the contract it also involves administration with a contract repository.
- An easy-to-access repository makes documents easy to search and find with just a few keywords.
- Ongoing contract is when you fulfill the obligations of your agreement up until the option or desire to renew.
- By the agreed-upon dates, you'll have to decide if a contract should be renewed or terminated.

Stages in Contract (Management & Renewal)

- This step is incredibly important renewing a contract isn't always beneficial to either party.
- This includes tasks like:
 - *Keeping track of contract milestones*
 - *Ensuring obligations are met after the contract is signed*
 - *Tracking changes or amendments to your contract*
 - *Executing periodic audits to ensure your contracts are compliant*

Stages in Contract (Reporting & Tracking)

- Especially if we're owning multiple contracts, it's difficult to continuously ensure things are on track.
- Missed deadlines for expirations, renewals, or goals are major contract management challenges and can lead to significant financial, legal, and procurement risk.
- Allowing contracts to auto-renew can also cost you compared to the opportunity to renegotiate.
- Knowing who accessed the records and made changes to them and when, is also important.

Stages in Contract (Reporting & Tracking)

- Issues like late payments, non-delivery of products, excessive changes to the contract, and other potential breaches need flagging for prompt attention.
- Contract management facilitates in-depth reporting.
- It also allows lawyers to set up tailored notification systems to complement a business's internal processes.

Placements

- Placements are the employment opportunities offered by staffing companies which can be ***permanent or contract basis***.
- Permanent placements are traditional, long-term positions that are hired on a full-time, permanent basis.
- Contract placements, on the other hand, are short-term positions that are hired on a temporary or project basis.
- For employers, permanent placements provide the advantage of having a known and reliable personnel presence in their organization.

Placements

- However, permanent placements also typically require more time and money to recruit and onboard a new employee.
- Contract placements, on the other hand, are more cost-effective for employers and often require less time and training to onboard a new employee.
- The biggest drawback to contract placements is that they are typically not available for long-term positions.

Placements

- When considering which type of placement is best for your organization, it is important to weigh the advantages and disadvantages of each.
- For example, if you require a long-term, reliable employee, a permanent placement is usually the best choice.
- However, if you need a short-term position filled quickly and cost-effectively, contract placements are the way to go.

Placements

- Commonly recognized types of contract placements:
 - Traditional contract staffing
 - Temp-to-direct hire conversions
 - Pay-rolling for non-recurited candidates.
 - Independent contractor to employee conversions.
 - Retiree re-staffing
 - Internships/co-operations

Typical Terms of Contract

- A contract term is any provision or term that forms part of a contract where every term provides a contractual obligation, which can lead to litigation if breached.
- Every contract a business enters will have critical terms that fall into various categories.
- The terms of an agreement can bind parties by law to meet a set of minimum obligations, which is why the groups involved will commonly go back and forth during negotiations on clauses and terms until both are satisfied.

Typical Terms of Contract

- Typically, contract terms can be defined into three categories:
 - Condition
 - Warranties
 - Innominate
- By categorizing contract terms into categories, it determines the available remedies if either party is ever in breach of the contract

Typical Terms of Contract

- ***Condition***

- If a term is a condition that is not followed, then the acting party will be in breach of contract, entitling the counterparty to terminate the contract and claim damages.

- ***Warranties***

- If a term is a warranty, the acting party can claim damages if the counterparty breaches the contract. However, they will not be able to terminate the agreement.

- ***Innominate***

- A term will be innominate if it is neither a condition nor a warranty. In the case of a breach of an innominate term, the type of action available to the acting party will depend on how severe the violation is.

Typical Terms of Contract

- The terms that have been specifically mentioned and agreed upon by the contracting parties at the negotiation stage are ***Expressed Terms***.
- These can either be in writing or agreed upon orally.
- ***Implied Terms*** are a set of default rules for contracts on points that are essentially "silent".
- These mandatory rules quietly operate in the background and can override the expressed terms.

Typical Terms of Contract

- But in many cases, there will be terms within a contract that have yet to be mentioned by either party involved.
- This does not mean they are not enforceable, and the agreement would not make sense without them.

Common contract terms:

- Confidentiality
- Termination
- Force Majeure
- Dispute
- Damages



Typical Terms of Contract

- ***Confidentiality***

- When a business enters a contract, there will be a significant exchange of information between the parties involved.
- It is standard practice that both parties keep sensitive information about their agreement confidential.

- ***Termination***

- The termination clause is found in every contract.
- This section lays out in which circumstances a party can terminate the agreement.
- In business, it is sometimes more financially beneficial to terminate the contract.

Typical Terms of Contract

- *Force Majeure*

- This phrase and clause translate to "greater force".
- This clause aims to protect parties from circumstances beyond anyone's control.
- For example, commonly, this clause refers to natural forces such as a hurricane or a pandemic, such as COVID-19.
- This clause is included to protect the party from being in breach of contract in the event of an unforeseeable disruption.

Typical Terms of Contract

- ***Dispute***

- Contracts form the basis of the terms and conditions in business relationships, and even well-written agreements can result in a dispute between the involved parties.
- The disputed clause clarifies the plans for any dispute resolution that arises.

- ***Damages***

- If either party breaches the contract, then damages are awarded to the innocent party or they are entitled to compensation.

Terms vs Clauses

Terms	Clauses
1. They form a basis of a contract. 2. They are broad agreement between the two parties outlining a contractual relationship 3. They establish the remedies available to either party in case of a breach of contract.	1. They are what a contract is composed of. 2. Contract clauses are blocks of text within an agreement that addresses and explains each part of the contract for the parties in more detail. 3. They are the conditions that must be met for the deal to go through successfully For example, a standard contract clause would be a termination clause. This clause will explain how either party can terminate the contract and what repercussions will be involved.

Contract Management

- It is the process of managing contracts, deliverables, deadlines, contract terms and conditions while ensuring customer satisfaction.
- Activities involved can be administrative and strategic - depending on who handles which stage.
- Effective contract management helps businesses to improve outcomes and realize maximum value from their agreements.
- Contracts should include agreement about how customer/supplier relationship is to be managed.

Contract Management

- Some terms of contract will relate to management of contract, for example
 - *Progress Reporting*
 - *Decision Points – could be linked to release of payments to the contractor*
 - *Variations to the contract, i.e. how are changes to requirements dealt with?*
 - *Acceptance Criteria*

Contract Acceptance

- When work is completed, customer needs to carry out acceptance testing.
- Contract may put a time limit to acceptance testing-customer must perform testing bf time expired.
- Part or all payment to the supplier should depend on acceptance testing



Managing People and Organization

Managing People

- There are three main concerns: staff selection, staff development and staff motivation.
- A project leader can encourage effective group working and decision making while giving purposeful leadership where needed.
- As part of the concern for the well-being of team members some attention must be paid to stress and other issues of health and safety.
- It is the most difficult areas in managing software development projects "Most managers are willing to acknowledge the idea that they've got more people worries than technical worries.
- But they seldom manage that way. ***One reason: technical experts become managers.***

Understanding Behavior

- People with practical experience of working on projects invariably identify the handling of people as one of the most important aspects of project management.
- What people want to know is whether the effective and sensitive management of staff comes only from experience or whether guidance can be usefully required.
- The field of social science known as organizational behavior (OB) helps to explain people's behavior and that tend to be structured .

Understanding Behavior

- It is then concluded that the effective and sensitive management of staff comes only from experience.
- So, the identification and handling of people as well as their behavior is a very much important aspect of project management.
- We should focus on discipline and organizational behavior, which can be done through following approaches:
 - *Positivist Approach*
 - *Interpretivist Approach*

Understanding Behavior

- ***Positivist Approach***

- It is based on development of system and discipline of organizational theory
- ***For example:*** “If A is the situation then B is likely to result”.
- Attempts are then made to observe behavior or to conduct experiments where variables for A and B are measured and a statistical relationship between the two variables is followed.
- Unlike physical science, it is rarely, if ever, the case that it can be said that B must always follow A.

Understanding Behavior

- ***Interpretivist Approach***

- It is more qualitative approach and view each and every consumption situations as unique and non-predictive.

Understanding Behavior

- A major problem is that in the real world there is guaranteed to be a very wide range of influences on a situation, many of which will not be apparent to the observer.
- It is therefore difficult to decide which set of research findings is relevant.
- A danger is that we end up with a set of hypothesis that are little better than superstitions.
- However, it is hoped that by examining these questions people can become more sensitive and thoughtful about the problems involved.

Organizational Behavior

- The roots of studies in OB can be traced back to work done in the late 19th and early 20th centuries by Frederick Taylor.
- By studying the way that manual workers did tasks, he attempted to work out the most productive way of doing these tasks.
- The workers were then trained to do the work in this way. Taylor had three basic objectives for training the worker:
 - *To select the best person for the job.*
 - *To instruct such people in the best methods.*
 - *To give incentives in the form of higher wages to the best workers.*
- Organizational behavior research found that the state of minds of people helps in better productivity.

Organizational Behavior

- Organizational behavior is a subset of management activities concerned with understanding, predicting and influencing individual behavior in an organizational setting.

Selecting Right Person for Job

- We need to select such person for the job who can affect the productivity of the project.
- Anyone who can communicate well and have a good experience and skills are the one who can influence the development of the institution.
- We need to distinguish between eligible (having the right qualifications) and suitable candidates (can do the job).
- The danger is employ someone who is eligible but not suitable

Selecting Right Person for Job

- The best situation is to employ someone who is suitable but not eligible! *For example, these are likely to be cheaper and to stay in the job.*
- So to select the right person for the job we can implement the selection process:
 - *Create of Job Specification.*
 - *Create a Job Holder Profile*
 - *Obtain Applicants*
 - *Examine CV*
 - *Interview*
 - *Other Procedure*

Selecting Right Person for Job

- ***Create of Job Specification:***

- Advice is needed, as there will be legal implications in an official document.
- However, formally or informally, the requirements of the job should be documented and agreed.

- ***Create a Job Holder Profile:***

- Using the job specification, a profile of the person needed to carry out the job is constructed.
- The qualities, qualifications, education and experience required would be listed.

Selecting Right Person for Job

- ***Obtain Applicants:***

- Typically, an advertisement would be placed, either within the organization or outside in the trade or local press.
- The job holder profile would be examined carefully to identify the medium most likely to reach the largest number of potential applicants at least cost.
- For example, if a specialist is needed it would make sense to advertise in the relevant specialist journal.
- By giving the salary, location, job scope and any essential qualifications, the applicants will be limited to the more realistic candidates.

Selecting Right Person for Job

- ***Examine CV:***

- These should be read carefully and compared to the job holder profile nothing is more annoying for all concerned than when people have CVs which clearly indicate that they are not eligible for the job and yet they are called for interview.

Selecting Right Person for Job

- ***Interview:***

- A number of different selection techniques can be tried, including aptitude tests, personality tests, and the examination of samples of previous work.
- All these methods must be related to specific qualities detailed in the job holder profile.
- Interviews are the most commonly used method and is better if there is more than one interview session with an applicant
- With each session there should not be more than two interviewers because a greater number reduces the possibility of follow-up questions and discussion.

Selecting Right Person for Job

- ***Other Procedure:***
 - References will need to be taken up where necessary, and a medical examination might be needed.

Instruction in Best Method

- When a new member of the team is recruited, the team leader will need to plan that person's induction into the team very carefully.
- Where a project is already well under way, this might not be easy.
- However, the effort should be made it should pay off eventually as the new recruit will become a fully effective member of the team more quickly.
- The team leader should also be aware of the need to assess continually the training needs of their team members.

Instruction in Best Method

- Some training can be provided by commercial training companies.
- Where money is tight, other sources of training should be considered but training should not be abandoned altogether even if it consists only of a team member's being told to find out about a new software tool and then demonstrating it to colleagues.
- The methods learnt need, of course, to be actually applied after training has been completed.
- Reviews and inspections should be done continuously which ensures all are on the track.

Motivation

- The process that account for an individual's intensity, direction and persistence of effort toward attaining a goal is called motivation.
- Motivation is a process that stands with a physiological or psychological deficiency or need that activates behavior or a drive that is aimed at goal or incentive.
- An important role of manager is to motivate the people working on project.
- It is complex issue but it appears to be different based on:
 - *Basic needs (food, sleep, clothing etc.)*
 - *Personal needs (respect, self-esteem).*
 - *Social needs (to be accepted as a part of a group)*

Motivation

- ***Characteristics of Motivation:***

- Motivation is a pervasive function and is performed by employees at all level
- It is always goal oriented and it involves effort to achieve goals.
- It is a psychological process and is concerned with individual's needs, motives, drives and other internal, status.
- It is complex and unpredictable due to the presence of individuals difference.
- It is situational because motivation differs person to person and time to time.
- It can be positive or negative. Positive motivation is rewarding whereas negative motivation is based on punishment.

Maslow's hierarchy of needs

A classic psychological framework from Abraham Maslow.

BiteSize Learning

Self-actualization

Finally, now our core needs are met, we're ready to pursue our deepest desire for **self-actualisation** - to become the most complete and authentic version of ourselves. With the fundamentals taken care of, we're free to act as we choose, aligned with our values and higher purpose.

Self esteem

Once we're secure in our social needs, our primal desires for **status and self esteem** come into play. We want to feel not only 'part of the group' - but an admired and respected part.

Social

If we feel safe, then our **social needs for love and belonging** come into play. We feel secure when we're part of a tribe we can trust and unworried about rejection or alienation.

Safety

If we feel OK in this exact moment, we then also want to **feel safe**: confident we can avoid pain, hunger and other dangers for the immediate future.

Physiological

At the very least, we don't want to be hungry, thirsty, exhausted, freezing, overheating, sick, or in pain.

Motivation

- Herzberg suggested two sets of factors affected job satisfaction
 - *Hygiene or maintenance factors* – make you dissatisfied if they are not right e.g. pay, working conditions
 - *Motivators* – make you feel the job is worthwhile e.g. a sense of achievement

HERZBERG'S THEORY OF MOTIVATION



SATISFACTION

- Achievement
- Recognition
- Growth
- Responsibility
- Advancement
- The work itself



DISSATISFACTION

- Security
- Status
- Salary
- Supervision
- Organizational policies
- Work environment



Expectancy Model of Motivation



Working in Groups

- Most software project development process is group activity as most of the projects cannot be completed by one person or working alone.
- Having discussed people as individuals, we move on to their place in groups.
- A key problem with major software projects is that they always involve working in groups, but as we have seen many people attracted to computer development find this difficult.

Working in Groups

- Formal groups can be subdivided into command groups, which are the departmental groupings that are seen on organization hierarchy diagrams and which reflect the formal management structure and task groups set up to deal with specific tasks.
- These call on people from different command groups and would typically be disbanded once the task has been completed.

Becoming a Team

- Simply throwing people together does not mean that they will immediately be able to work together as a team.
- Group feelings develop over a period of time.
- One suggestion is that teams go through five basic stages of development:
 - *Forming*
 - *Storming*
 - *Norming*
 - *Performing*
 - *Adjourning*

Becoming a Team

- ***Forming:***

- The forming stage involves a period of orientation and getting acquainted.
- Uncertainty is high during this stage, and people are looking for leadership and authority.
- A member who asserts authority or is knowledgeable may be looked to take control.
- Team members are asking such questions as “What does the team offer me?” “What is expected of me?” “Will I fit in?”
- Most interactions are social as members get to know each other.

Becoming a Team

- ***Storming:***

- The storming stage is the most difficult and critical stage to pass through.
- It is a period marked by conflict and competition as individual personalities emerge.
- Team performance may actually decrease in this stage because team members may disagree on team goals, and subgroups and groups may form around strong personalities or areas of agreement.
- To get through this stage, members must work to overcome obstacles, to accept individual differences, and to work through conflicting ideas on team tasks and goals
- Failure to address conflicts may result in long-term problems.

Becoming a Team

- *Norming*

- If teams get through the storming stage, conflict is resolved and some degree of unity emerges.
- In the norming stage, consensus develops around who the leader or leaders are, and individual member's roles.
- Interpersonal differences begin to be resolved, and a sense of cohesion and unity emerges.
- Team performance increases during this stage as members learn to cooperate and begin to focus on team goals.
- However, the harmony is precarious, and if disagreements re-emerge the team can slide back into storming.

Becoming a Team

- *Performing:*

- In the performing stage, consensus and cooperation have been well-established and the team is mature, organized, and well-functioning.
- There is a clear and stable structure, and members are committed to the team's mission.
- Problems and conflicts still emerge, but they are dealt with constructively.
- The team is focused on problem solving and meeting team goals.

Becoming a Team

- *Adjourning:*

- In the adjourning stage, most of the team's goals have been accomplished.
- The emphasis is on wrapping up final tasks and documenting the effort and results.
- As the work load is diminished, individual members may be reassigned to other teams, and the team scatters.
- There may be regret as the team ends, so a ceremonial acknowledgement of the work and success of the team can be helpful.
- If the team is a standing committee with ongoing responsibility, members may be replaced by new people and the team can go back to a forming or storming stage and repeat the development process.

Decision Making

- A decision-making process is a series of steps one or more individuals take to determine the best option or course of action to address a specific problem or situation.
- Often, managers and executives use the process to plan how to carry out business initiatives or set specific actions in motion.
- Decision Making can be:
 - *Structured*
 - *Un-structured*

Decision Making

- ***Structured:***

- Goals are defined in the structured decision.
- Information is obtainable and manageable.
- Appear in a well-defined context and procedures are known.
- The cost of taking such a decision is not as high as that of an unstructured one.
- They are essentially repetitive, routine, and involve a defined.
- Structured decisions are the programmable decisions and they are preplanned *for example the payroll for employees.*

Decision Making

- *Un-structured*

- Outputs are uncertain in the unstructured decision.
- The required information and resource are hard to assess.
- Appear in a unique context in the unstructured decision.
- The cost of taking such decisions is quite high, compared to structured decisions.
- There are no predefined procedures available to solve.
- *Decision making on personal development* is example of un-structured decision making.

Decision Making

- ***Obstacles to Decision making:***
 - It is time consuming.
 - It can mixture up conflicts within the group.
 - Decisions can be unduly influenced by dominant personalities.
- Conflict can, in fact, be less than might be expected.
- People will modify their personal judgments to conform to group norms, common attitudes developed by a group over time.

Leadership

- Leadership is generally taken to mean the ability to influence others in a group to act in a particular way to achieve group goals.
- A leader is not necessarily a good manager or vice versa as managers have other roles such as organizing, planning and controlling.
- Leadership is based on the idea of authority or power although leaders do not necessarily have much formal authority.

Leadership

- Power may come either from the person's position (position power), from the person's individual qualities (personal power) or may be a mixture of the two.
- Position power has been further analyzed into:
 - ***Coercive Power:*** The ability to force someone to do something by threatening punishment
 - ***Connection Power:*** Based on having access to those who have power
 - ***Legitimate Power:*** Based on a person's title conferring a special status
 - ***Reward Power:*** The holder can give rewards to those who carry out tasks to his or her satisfaction

Leadership

- Personal power has been further analyzed into:
 - ***Expert Power***: Comes from being the person who is able to do a specialized task
 - ***Information Power***: The holder has exclusive access to information.
 - ***Referent Power***: Based on the personal attractiveness of the leader

Measuring Leadership

- Leadership styles can be measured on two axes: Directive vs. Permissive and Autocratic vs. Democratic.
 - ***Directive Autocrat:*** Makes decisions alone; close supervision of implementation
 - ***Permissive Autocrat:*** Makes decision alone, subordinates have autonomy in implementation
 - ***Directive Democrat:*** Makes decisions participatively, close supervision of implementation
 - ***Permissive Democrat:*** Makes a decision participatively, subordinates have autonomy in implementation

Organizational Structure

- Any operating organization should have its own structure in order to operate efficiently
- The organizational structure is a hierarchy of people and its functions.
- The organizational structure of an organization tells you the character of an organization and the values it believes in.
- Therefore, when you do business with an organization or getting into a new job in an organization, it is always a great idea to get to know and understand their organizational structure.

Organizational Structure

- Depending on the organizational values and the nature of the business, organizations tend to adopt one of the suitable structures for management purposes.
- Although the organization follows a particular structure, there can be departments and teams following some other organizational structure in exceptional cases.
- Here we focus on these two structures:
 - Functional Structure
 - Matrix Structure

Functional Structure

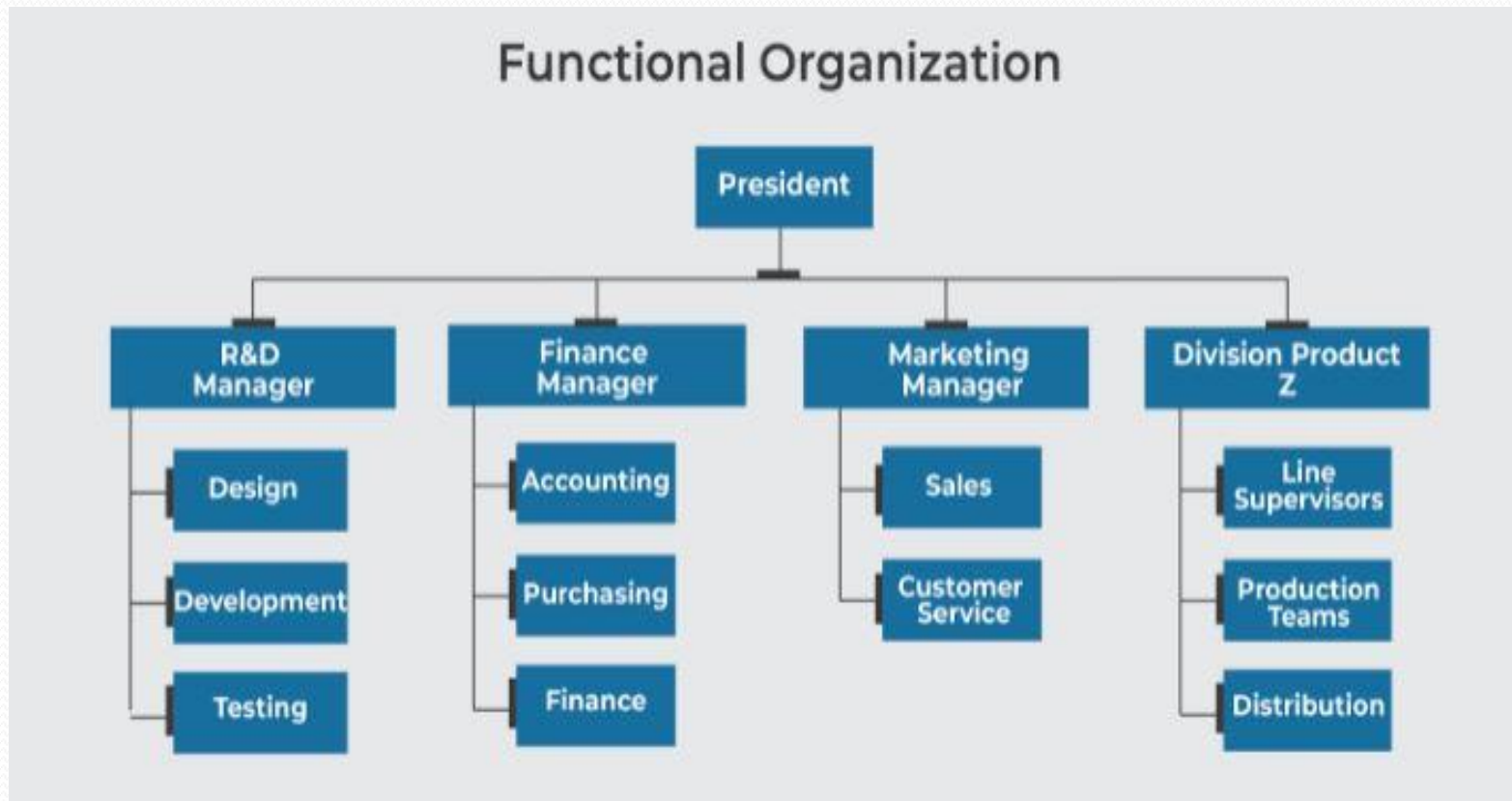
- The organization is divided into segments based on the functions when managing.
- This allows the organization to enhance the efficiencies of these functional groups.
- Functional structures appear to be successful in large organization that produces high volumes of products at low costs.
- The low cost can be achieved by such companies due to the efficiencies within functional groups.

Functional Structure

- Here you classify workers according to their functional roles and department.
- Some of the general departments under this are
 - Finance
 - HR
 - Sales
 - Customer service
 - Supply Chain, etc.
- This type of organization is suitable for manufacturing or engineering companies.

Functional Structure

- It supports ongoing operations and practices for producing standard products.



Functional Structure

- **Advantages**

- It groups employees based on functional skills for a higher degree of performance.
- Employees have experience in the same field, resulting in higher output. Also, they deliver high-quality services and results.
- Accountability is evident, as the roles and responsibilities are clearly defined.
- Hierarchy is visible and no need for multiple reporting
- No duplication of work as each department is different. Also, the job description is clear.

Functional Structure

- **Disadvantages**

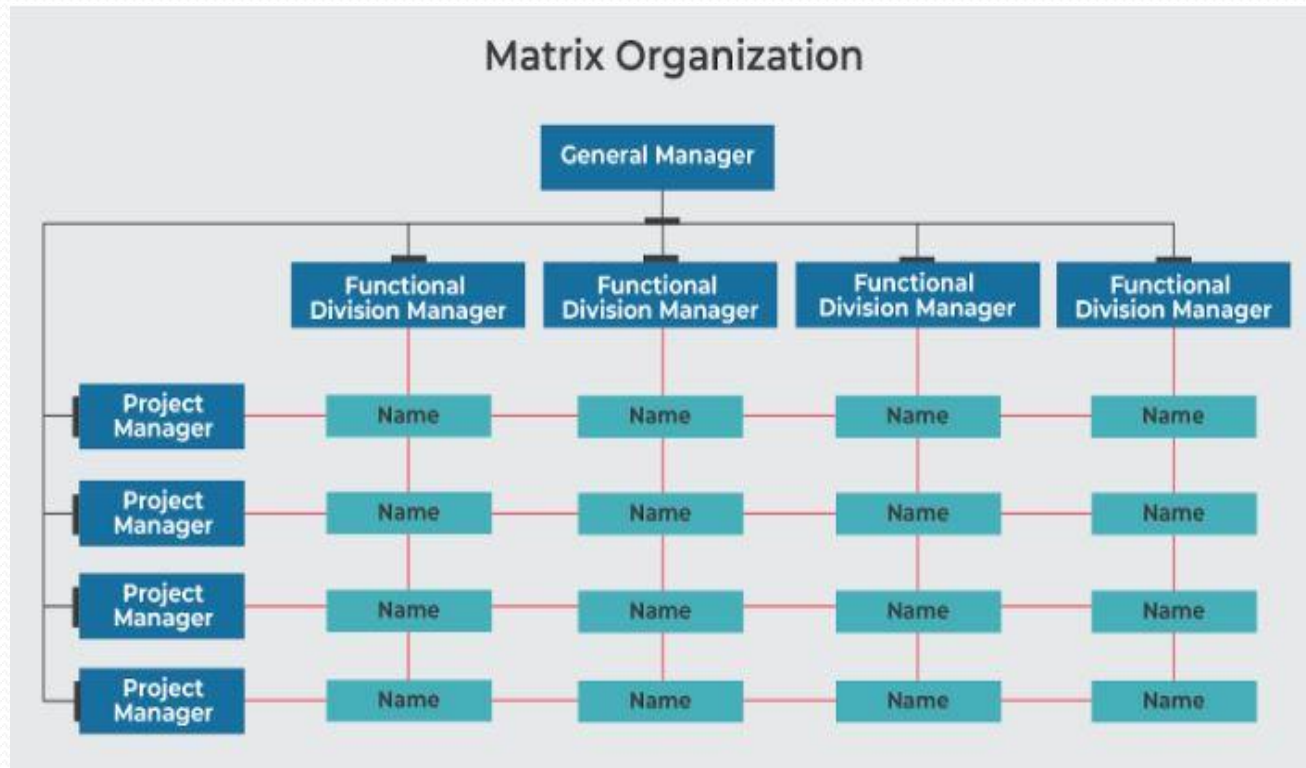
- Employees get bored from the routine and lose enthusiasm.
- It limits the management skills of functional managers. Hence, they face restrictions on their growth path and remain specialists. So they're not prepared for top management posts.
- Departments are more concerned with their departmental goals, hence may be less responsive to the organization's overall objectives.
- Hiring costs are too high as high-skilled employees cost more.
- It causes conflicts in making critical decisions as a result of bureaucratic hierarchy.

Matrix Structure

- When it comes to matrix structure, the organization places the employees based on the function and the product.
- The matrix structure gives the best of the both worlds of functional and divisional structures.
- In this type of an organization, the company uses teams to complete tasks.
- The teams are formed based on the functions they belong to (ex: software engineers) and product they are involved in (ex: Project A).

Matrix Structure

- This way, there are many teams in this organization such as software engineers of project A, software engineers of project B, QA engineers of project A, etc.



Matrix Structure

- **Advantages**

- It helps in sharing resources efficiently
- Decision making is balanced and flexible
- Staff members can communicate with each other across boundaries
- Pleasant environment
- It has a clear career graph and job security; hence, members would be more loyal to the organization

Matrix Structure

- **Disadvantages**

- The dual reporting structure add confusion and results in conflicts.
- Create issues when there is no co-ordination between functional and project managers.
- Resources may be under-utilized if you don't assign them with skill-related tasks.
- Costly to maintain as it has many managers.
- You need to maintain resources throughout the project, no matter how long it takes.

Organizational Structure Influence in Projects

- A company with shared goals, strong leadership and hard-working employees can still fail without organizational structure.
- While organization is necessary for success, the structure can influence the project management process.
- A company's organizational structure may dictate the level of project management, who makes ultimate project decisions, the communication of project goals and tasks and how the project manager works with his team.

Organizational Structure Influence in Projects

Project Characteristics	Organizational Structure Type				
	Functional	Matrix			Project
		<i>Weak Matrix</i>	<i>Balanced Matrix</i>	<i>Strong Matrix</i>	
Project manager's authority	Little or none	Limited	Low to Moderate	Moderate to high	High to almost total
Percent of performing organization's personnel assigned full-time to project work	Virtually none	0-25%	15-60%	50-95%	85-100%
Who controls the project budget	Functional manager	Functional manager	Mixed	Project manager	Project manager
Project manager's role	Part-time	Part-time	Full-time	Full-time	Full-time
Common title for project manager's role	Project Coordinator/ Project Leader	Project Coordinator/ Project Leader	Project Manager/ Project Officer	Project Manager/ Program Manager	Project Manager/ Program Manager
Project management administrative staff	Part-time	Part-time	Part-time	Full-time	Full-time

PMBOK® Guide, 2000, 19, and PMBOK® Guide 2004, 28.

Assignment-III

- Explain Maslow's hierarchy of needs.
- How you can select the right person for the job? Explain.
- What are the roles of Software Engineer in a project? Explain.
- What are the popular organizational structure adopted by Nepalese companies. Explain
- Explain the process of becoming a mature team within an organization.
- What are the popular contract types preferred by the organizations? Also explain stages of contract lifecycle.
- Describe the typical terms of a contract.



End of Chapter 7

Thank You !!!!